

Understanding Group Engagement in Online Peer Support Groups: Influence of Structured Elements and Organic Interactions

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Abstract

Online peer support groups (OPSGs) have been widely used in health and wellbeing interventions, but their effectiveness varies, influenced by the level of group engagement and the quality of interactions. This study explores the impact of structured elements, such as prompts and tasks, incorporated into OPSG design on peer support quality and group engagement. We analyse chat messages from two OPSGs aimed at reducing workplace sitting through exercise breaks. Despite identical study designs and similar participant demographics, the two OPSGs exhibited significant differences in group engagement, underscoring the importance of understanding group composition, organic interactions, and other factors influencing engagement. The findings emphasise the need for thoughtful OPSG design, including developing structured elements, to foster communication and improve the quality of interactions. Based on our findings, we discuss strategies to promote supportive exchange, focusing on the role of prompts, group cohesion, and social presence to enhance the effectiveness of OPSGs.

CCS Concepts

• **Human-centered computing** → **Social network analysis**; *Social networking sites*; *Social media*; *Social networks*; Empirical studies in collaborative and social computing.

Keywords

Peer Support Groups, Healthcare, Wellbeing, Social Support, Mobile Health, Social Media

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1 Introduction

Peer support plays a crucial role in helping individuals navigate challenging situations, offering empathy, validation, and practical advice through shared experiences beyond the scope of professionals [31]. With the rise of mobile phones and social media, online peer support groups (OPSGs) have become popular for managing physical and psychological health conditions, offering benefits such as reducing stigma, decreasing loneliness and anxiety, enhancing wellbeing, and providing opportunities for self-expression [22, 25, 58]. Research shows that promoting connection with like-minded people can foster autonomous motivation for behaviour change [8, 14]. OPSGs can influence members' health behaviours through social support, peer pressure, and information sharing [2, 14]. However, their effectiveness depends on various factors, including perceived relatedness, motivation, and the quality of interactions [2, 14].

OPSGs come in various forms, ranging from naturally occurring groups to those intentionally created and designed for a specific purpose. Research focus also varies; studies on the former often conduct the content and network analysis [11, 44], while studies on the latter typically evaluate their effectiveness in achieving health and wellbeing outcomes [30], leaving a gap in understanding how groups' structured elements, such as educational materials, tasks, and prompts, affect peer support quality and group engagement.

This paper aims to address this gap by analysing chat messages from two OPSGs designed to reduce prolonged workplace sitting through exercise breaks [34]. Our contribution includes the development of a coding scheme informed by established frameworks such as the Turning Toward Versus Turning Away Coding System (Turning System) [16] and the Assessment of Social Presence

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(ASP) [41]. We used these to analyse the chat engagement, focusing on differences in message activity across categories. Additionally, we explored differences in requested and provided support, using the Social Support Behaviour Code (SSBC) framework [13], and discussed how the support exchange could be promoted.

2 Related Work

2.1 Self-organised vs Organised Groups

Self-organised groups were defined as naturally occurring online communities where participants come together voluntarily to share experiences, provide support, and exchange information. These groups often arise on forums, social media, or dedicated platforms [54]. These groups have become popular platforms for behaviour change, such as smoking cessation and weight management [27, 44, 47] and supporting people with chronic health conditions, such as diabetes, HIV, asthma, arthritis and cardiovascular disease [20, 23, 24, 42, 44, 46, 55].

Research on self-organised OPSGs has employed various approaches [45], including examining the structure of social networks to understand behaviour patterns [19]; exploring topics, themes, opinions, or sentiments within chats [3, 35, 46]; combining content and network analysis [57]; and examining the intent and context of language use, focusing on how speech acts like promises, questions, and suggestions shape communication dynamics [44]. The results have highlighted the ability of these groups to foster organic connections among participants with shared experiences, demonstrating their effectiveness in providing support and education and promoting self-management [23, 42, 55].

By *organised groups*, we understand structured environments created and facilitated by researchers for specific interventions. These OPSGs can be placed on comprehensive websites, purposely designed platforms or social media [25]. These groups involve pre-recruited participants who interact within a framework designed to achieve targeted outcomes or change behaviour [39]. For example, researchers provide educational materials, set tasks and provide prompts to moderate discussions, ensuring participants stay on track [29, 30, 40, 52]. The organised groups are different from the comprehensive online support services, which provide an assortment of options, such as online encyclopedias, question-and-answer capabilities, and online discussion groups, but don't require participants to use them or follow specific instructions [54]. Researchers usually examine whether integrating OPSGs into interventions can lead to health behaviour changes, e.g., increase physical activity [29] and reduce smoking relapse [10].

2.2 Support Exchange Promotion

OPSG interventions are interactive, involving exchanges between the research team and group members and among group members themselves [39]. However, research shows that OPSG members often fail to respond to messages from support seekers, limiting group effectiveness and causing feelings of isolation [1, 51]. Balanced participation can enhance discussion quality, member satisfaction, and group dynamics, yet low engagement from some members is common [15]. Interest often wanes over time due to delayed responses or loss of motivation [40], and even trained peer leaders

face challenges with sustained involvement, often discontinuing their participation over time [49].

Group moderators can take an active role by guiding discussions with prompts [39, 43]. The effective prompts can engage group members in study components and promote group cohesion [39], which refers to the sense of belonging and acceptance to the group and is linked to better attendance, interpersonal support, and better outcomes [6]. Moreover, research suggests that some types of group interactions, e.g., sharing personal experiences and expressing gratitude, can effectively facilitate social support exchanges among group members [12].

The challenges of human communication, combined with advances in large language models [38, 50], have led researchers to investigate the use of conversational agents (CAs) in OPSGs, as CAs offer scalable, timely interventions, enhancing support without relying solely on human moderators [15]. For example, research comparing AI-based CA and human social support for distress reduction showed that the perceived CA competence can ease distress indirectly when human warmth can do it directly [32].

3 Methods

To explore the balance between structured elements and organic interactions and how this influences peer support, our analysis focused on data from intervention groups in Cohorts 2 and 3¹ of the randomised controlled trial Move Often eVery Day (MOV'D) which aimed to interrupt prolonged workplace sitting with exercise breaks [34]. The group chat had a unique structure: intervention participants were randomized into pairs of "buddies" [51] for mutual support. All communication was held in the group chat to ensure everyone could benefit from group activities, even those who didn't get a good match with their buddy [40]. The research team was in the chat to answer technical questions and send study materials: behaviour change technique (BCT) videos, exercise snack videos (called a "snackivity" in the study) and intervention prompts. Participants were expected to interact with the study materials and share their weekly exercise goals and everyday progress. All study components with their description are listed in Appendix A.

Compared to Cohort 2 (12 participants), Cohort 3 (13 participants) was slightly older (mean age 41.62 (SD=10.68) vs 37.08 (SD=9.13)) with a predominantly female population (92% female vs 80%), as reported by Uetova et al. [51].

3.1 Data Collection and Analysis

The chat logs² were retrieved from the GroupMe app and anonymised by replacing participants' names with pseudonyms.

The analysis consists of five parts. In Section 4.1, we analysed the differences in group interaction patterns between Cohort 2 and 3, focusing on the frequency and number of message exchanges. In Section 4.2, we developed a coding scheme using both inductive and deductive thematic analysis approaches [5]. This process was based on the premise that communication in OPSGs has similar patterns as in-person groups where common themes include emotional

¹The intervention groups in Cohorts 2 and 3 both used GroupMe as their communication channel, whereas Cohort 1 used Twitter [34].

²Some participants initially used private chats as the main form of communication with their buddy. However, after one week, all messages were redirected to the group chat. These private chats were excluded from the analysis.

Table 1: Example of segmenting and annotating a single message.

Message's Part	Category	Study Component
@P73	Social Presence	Not Applicable
My goal for this week is 3 snack-tivities.	Action Planning	Goal Setting
I watched the Mind-Plan Creation mode.	Interaction Confirmation	BCT video
The video is suggesting we come up with many ideas to get ourselves moving. We will try each idea and see which one works best.	Information Sharing	BCT video
One of my ideas is to set a timer for every 1.5 hour of sitting, I will get up and do a dance snacktivity which makes me feel good.	Action Planning	BCT video
What is your goal for this week?	Unprompted Support	Goal setting

support, information, and self-disclosure, with participants able to request and provide these elements [56]. The coding scheme incorporated the unique structure of the MOV'D group chats, recurring concepts, and established systems like the Turning System [16] and the ASP [41]. To accommodate diverse messaging styles—where some participants sent multiple messages on different topics, and others combined everything into a single message—we segmented each message into distinct parts and annotated each part with a single label, see an example in Table 1. In Section 4.3, we analysed the engagement using our coding scheme with a focus on differences in message activity across categories and the influence of intervention components on group dynamics. In Section 4.4, we explored differences in requested and provided support types based on the SSBC framework [13] and in Section 4.5, we examined changes in engagement over time.

4 Results

4.1 Interaction Patterns

Despite the similar number of participants (12 people in Cohort 2 and 13 people in Cohort 3), demographics (predominantly female population), and identical intervention components and study design, Cohorts 2 and 3 varied significantly in the total number of messages exchanged, the frequency of interactions with other participants, and the overall level of activity, see Table 2. Participants in Cohort 3 sent more messages and engaged more broadly, communicating with more unique recipients and receiving more messages than Cohort 2. This suggests that the level of engagement may be influenced more by other factors, e.g., differences in group composition or organic interactions, than by the OPSG design. Reflecting the data in Table 2, a few talkative participants in each cohort sent significantly more messages, leading most other participants to receive more messages than they sent. This aligns with findings by Nonnecke and Preece [37], who observed that a

Table 2: Messaging and interaction patterns per participant in Cohort 2 and Cohort 3.

	Cohort 2 (N = 12)	Cohort 3 (N = 13)
Participant-generated messages ¹ , total, mean (SD; range)	147, 12.3 (7.4; 0-25)	355, 27.3 (13.3; 1-50)
Messages sent to peers, mean (SD; range)	1.8 (2.1; 0-6)	14.0 (8.3; 0-26)
Messages sent to the research team, mean (SD; range)	0 (0; 0-0)	1.0 (0.9; 0-2)
Messages sent to the group ² , mean (SD; range)	10.6 (6.4; 0-23)	13.4 (6.7; 1-26)
Unique recipients ³ , mean (SD; range)	1.0 (1.2; 0-4)	3.2 (1.8; 0-6)
Messages received from peers, mean (SD; range)	1.8 (2.2; 0-6)	15.6 (7.6; 0-26)
Unique senders, mean (SD; range)	1.0 (1.0; 0-3)	3.2 (2.2; 0-8)

¹ Without messages from the research team.

² The "group" means the messages were sent without specifying the recipient.

³ Excluding messages to the research team and the group.

small percentage of participants often account for the majority of postings, leading to an imbalance in group interaction. In Cohort 2, one person never sent a message, and five never received one. In Cohort 3, all participants sent at least one message, and only one participant never received a message. Moreover, participants communicated beyond their buddy pairs, interacting with others in the group. In contrast, Cohort 2 had one participant who initiated and received most of the communication, as shown in Appendix B. The conversational threads were short, and most contained only one or two messages. Only one thread reached six messages, and there were no threads longer than that.

4.2 Coding Scheme

We categorised messages as Prompted and Unprompted to reflect the unique structure of organised groups, see Table 3³. Prompted messages, which were encouraged by the research team through study components, fell into three categories: Interaction Confirmation, Information Sharing, and Action Planning. Unprompted messages, initiated voluntarily by participants, are categorized into six types based on the Turning System [16]. These categories are divided into two main groups: Bids and Responses. Bids represent attempts by participants to interact with others and include two types: Bids for Interaction, which encourage interaction using jokes, questions, or comments, and Unprompted Support, which provides unprompted inquiries or helpful information. Responses reflect reactions to the Bids and are classified as Turning Toward, indicating a positive, supportive response; Turning Against, denoting a negative or confrontational response; and Turning Away, signifying a lack of response. Some of the Unprompted message types can be further classified by support function, requesting or providing, and support type based on the SSBC [13], see Section 4.4. In addition to the Prompted and Unprompted types of messages, we included the

³We did not include categories that we did not analyze, such as corrections of typos, messages from the research team and system messages, e.g. when a new participant joins the group.

Table 3: Categories of our coding scheme.

Type	Systems	Category	Definition	Examples
Prompted	Study component	Interaction Confirmation	Message indicating participant engagement with study components, such as providing brief responses to prompts, reporting on snackactivities, or confirming having watched a behaviour change video.	<i>"I managed to fit in 2 bouts of today's suggested snackactivity doing the mountain climbers from the floor."</i>
		Information Sharing	Messages where participants shared information related to study components, such as providing suggestions for prompt tasks, teaching buddies behaviour change techniques from the videos, or sharing which provided snackactivity video they preferred to do.	<i>"I watched the Mind-Plan Creation mode. The video is suggesting we come up with many ideas to get ourselves moving. We will try each idea and see which one works best."</i>
		Action Planning	Message where participants outlined their strategies for implementing newly acquired knowledge to aid habit formation or shared their planned snackactivities, type and/or number, for the day or the upcoming week.	<i>"One of my ideas is to set a timer for every 1.5 hour of sitting, I will get up and do a dance snackactivity which makes me feel good."</i>
Unprompted	Bid	Bids for Interaction	Messages aiming to initiate or encourage interaction, including self-disclosure, jokes, comments, and questions, as well as requesting support.	<i>"Love the video and does look like fun. But is there an easier one for those of us who are move challenged"</i>
		Unprompted Support	Messages where participants provide support by checking in on buddies' study-related activities or offering helpful information without being prompted.	<i>"What is your goal for this week?" "I did this video this morning for all the folks looking for more dance workouts https://youtu.be/..."</i>
	Response	Turning Toward	Messages that provide positive, supportive responses, including compliments, gratitude and emotional reactions (e.g., "nice!").	<i>"worst nightmare! I'm sure people would have been so encouraging though!" "I am impressed by your creativity: Simon says, steps ups during hall duty..."</i>
		Turning Against	Messages expressing irritability or hostility, contrasting with supportive messages.	<i>No such messages were found in our dataset.</i>
Other	ASP	Turning Away	Messages where engagement attempts go unanswered, or a new topic is introduced without addressing the initial message, reflecting a lack of support.	<i>The absence of messages itself serves as the example, making it impossible to provide a tangible illustration of no message.</i>
		Social Presence	Messages serving social functions, like maintaining connection and facilitating communication [33], include phatic expressions (e.g., greeting, farewell, interjection, apologies), vocatives (i.e., referring to participants by name), and expressions of emotions (e.g., emoji, unconventional use of punctuation and capitalization).	<i>"Hi, everyone! 🍷" "@P81 TGIF! We made it to Friday." "Wishing you all the best of luck on your respective health journeys!" "oh man" "Looking forward to getting back into it today!!!"</i>

Social Presence category based on the ASP [41]. We chose several cues that fit our data well and don't overlap with other categories, such as expression of emotions, vocatives and phatic expressions. The Social Presence category reflects the group's social environment, ranging from cold and pragmatic (low usage of the cues) to warm and collegial (high usage).

4.3 Differences Between Cohorts Based on the Coding Scheme and Study Components

In Cohort 2, 147 messages were segmented into 239 parts based on our coding scheme (see an example in Table 1), while in Cohort 3, 355 messages were divided into 923 parts. The distribution of the message parts across categories and study components is shown in Table 4. The most common category for both cohorts was Interaction Confirmation, representing minimal but expected interactions, where participants adhered to the study requirements without extending beyond what was prompted. Cohort 3 demonstrated substantially higher engagement across most categories and study components. There was minimal variance in interaction levels observed among participants from both cohorts engaging with Prompts and only a slight discrepancy in engagement with BCT Videos, primarily related to Action Planning. This suggests that interaction with Prompts and BCT Videos may not influence participants' engagement in the group chat. However, participants in Cohort 3 demonstrated significantly higher activity in sending Snackactivity-related messages, tenfold more than Cohort 2, and Social Presence, seven times more frequently. Notably, Cohort 2 sent significantly fewer responses, Turning Toward, compared to Cohort

3, further highlighting differences in the groups' engagement and quality of interactions.

4.4 Types of Provided and Requested Support

The Bid for Interaction (requesting support in bid), Unprompted Support (providing support in bid) and Turning Toward (providing support in response) categories fell into four types of social support [13]: emotional support (e.g., encouragement), informational support (e.g., advice), esteem support (e.g., compliment), and network support (e.g., presence), see Table 5. Tangible support, which involves providing or offering physical assistance, was not present in our data. Some of the messages in these three categories did not provide or request any support and were marked as "No support". More details and examples are given in Appendix C.

Neither cohort requested network support. Participants most often sought esteem support (66.7% in Cohort 2 and 39.3% in Cohort 3) and emotional support (15.2% and 31.7%, respectively). Cohort 2 sent only two Unprompted Support messages, network and informational support, while Cohort 3 sent 64 such messages across all four categories, 85.9% of which were network support. In the Turning Toward messages, 81.8% of Cohort 2's responses provided esteem support, while the rest 18.2% provided informational support. In contrast, Cohort 3's responses spanned all four categories, with esteem support being the most prevalent (41.3%) and a notable proportion falling into emotional support (26.9%).

Table 4: Distribution of the number of message's parts across categories and study components¹.

Type	Category	Prompts		BCT Videos		Snackactivities ²		Other ³		Total	
		C2 ⁴	C3 ⁵	C2	C3	C2	C3	C2	C3	C2	C3
Prompted	Interaction Confirmation	60	64	23	36	9	101	X		92	201
	Information Sharing	15	13	22	23	0	1			37	37
	Action Planning	3	2	5	35	16	53			24	90
Unprompted	Bids for Interaction	20	18	3	13	9	95	1	19	33	145
	Unprompted Support	0	4	0	1	1	52	1	7	2	64
	Turning Toward	8	9	0	14	1	55	2	26	11	104
Social Presence								40	288	40	282
Total		106	110	53	122	36	357	44	340	239	923

¹ We did not provide numerical data for the Turning Away category, as sometimes determining whether a message requires a response is challenging.

² Snackactivities include videos, daily progress and goal setting.

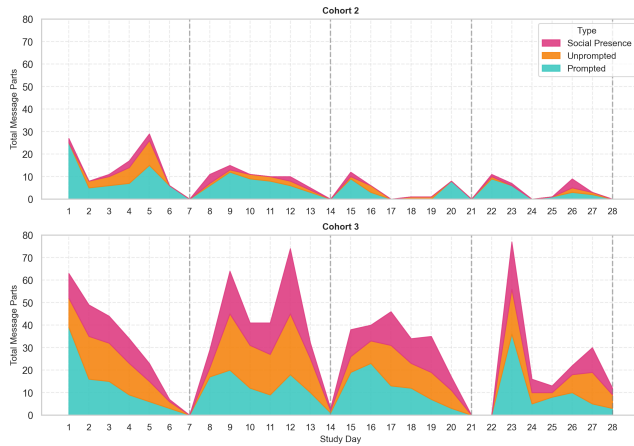
³ Other - messages not related to study components.

⁴ C2 - Cohort 2.

⁵ C3 - Cohort 3.

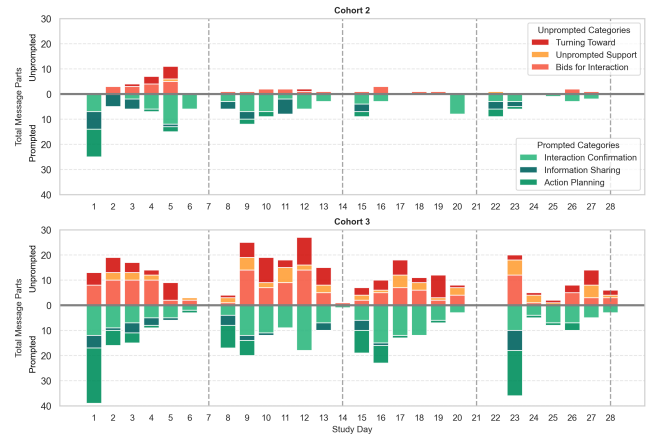
Table 5: Types of requested and provided social support.

Type	Requested Support		Provided Support			
	Bid for Interaction		Unprompted Support		Turning Toward	
	Cohort 2	Cohort 3	Cohort 2	Cohort 3	Cohort 2	Cohort 3
Esteem Support	66.7% (22 / 33)	39.3% (57 / 145)	0	4.7% (3 / 64)	81.8% (9 / 11)	41.3% (43 / 104)
Informational Support	3.0% (1 / 33)	4.8% (7 / 145)	50.0% (1 / 2)	3.1% (2 / 64)	18.2% (2 / 11)	5.8% (6 / 104)
Network Support	0	0	50.0% (1 / 2)	85.9% (55 / 64)	0	5.8% (6 / 104)
Emotional Support	15.2% (5 / 33)	31.7% (46 / 145)	0	6.3% (4 / 64)	0	26.9% (28 / 104)
No support	15.2% (5 / 33)	24.1% (35 / 145)	0	0	0	20.2% (21 / 104)

**Figure 1: Change over time in message parts by types. Vertical lines indicate the end of each week, with weeks running from Sunday to Saturday.**

4.5 Change over Time

OPSGs often experience fluctuations in participation, with engagement peaking during certain phases and declining over time. To examine these patterns, we analysed four weeks of data using our coding scheme. Figure 1 illustrates the change over time in Social Presence, Prompted and Unprompted messages. Participants in Cohort 3 not only regularly sent Prompted messages, showing engagement

**Figure 2: Change over time in message parts by categories. Vertical lines indicate the end of each week, with weeks running from Sunday to Saturday.**

with structured elements, but also actively contributed organically, sending more Unprompted messages and Social Presence cues compared to participants in Cohort 2. Moreover, Cohort 3 maintained a stable level of engagement throughout all four weeks, whereas Cohort 2's participation noticeably declined after the second week, with only occasional attempts to engage, primarily with structured elements via Prompted messages. Figure 2 provides a more detailed

view, focusing on six message categories—three Prompted (Interaction Confirmation, Information Sharing and Action Planning) and three Unprompted (Bids for Interaction, Unprompted Support, and Turning Toward). Interaction Confirmation was the most prevalent and consistent category across all four weeks in both cohorts. Participants in Cohort 3 regularly sent all categories of Unprompted messages, while participants in Cohort 2 barely sent any, with the majority being sent during the first week. This suggests a higher level of group engagement and better quality of interactions in Cohort 3, where participants were not only engaged with structured elements but also initiated and sustained discussions, fostering a stronger sense of connection among participants. In addition to this, we observed a change in the message parts related to study components, see Appendix A.

5 Discussion

This study explored communication patterns in organised OPSGs and examined the potential role of structured elements. While limited by sample size, the findings highlight the complexity of online group interactions and include:

Structured Elements Alone May Not Enhance Engagement.

Our findings reveal that, despite identical study designs and similar participant demographics, the two OPSGs exhibited significant differences in their levels of engagement. This implies that group engagement and quality of interactions may be shaped more by factors such as group composition or organic interactions than by the OPSG design. Minimal variance in interaction levels with Prompts and BCT Videos suggests that structured elements alone do not significantly influence group engagement. The critical factors appear to be the Snacktivity-related messages and Social Presence. Snacktivity-related messages include sharing everyday progress and challenges, as participants naturally experience a range of emotions, such as joy from achieving their goal, worry about others' perceptions while exercising at work or feeling out of shape. This encourages sharing thoughts and motivates providing mutual support, in accordance with Huang et al. [21] findings indicating that people perceived conversations as most necessary for addressing social and personal matters. Research shows that participants of peer support programs like it when others check in with them and highly value supporter's presence, "just being there" [18]. The more people share their stories, the more others reciprocate, increasing mutual understanding, trust, support-seeking behaviours, and satisfaction [41]. Unlike in-person groups, asynchronous web-based groups lack nonverbal cues (e.g., facial expressions), which reduces emotional connection [39]. For group cohesion, it is essential to actively engage group members and cue them to share their experiences [39].

Social Presence Influences Group Atmosphere. Social Presence cues help compensate for the lack of nonverbal communication by shaping the group's social environment. Low social presence suggests a cold, impersonal atmosphere focused on pragmatic interactions, while high social presence fosters a warm, collegial environment where participants feel a strong sense of belonging [41]. This sense of belonging motivates participants to value group interactions, encouraging them to share their thoughts and engage with others' posts [28, 41].

Support Types Reflect Group Needs and Dynamics. Previous research highlights the importance of social support for well-being and various health conditions in online settings [36, 48, 59]. The prevalent type of support can depend on the group's characteristics, such as its purpose, participants, and overall structure. For instance, participants of diabetes OPSG primarily sought and provided informational support [42]; in contrast, cancer caregivers primarily requested and received emotional support [4]. The support types in MOV'D group chats reflected their purpose of habit adoption, idea exchange, and learning behaviour change techniques. Esteem support, including validation and compliments, was most common, followed by emotional support. Information was rarely requested or provided due to its availability from the research team. Network support, while never requested, was frequently offered in Cohort 3, fostering connections and group cohesion. Cohort 3 demonstrated greater diversity in support types and higher peer support quality, indicating that some groups promote more supportive environments while others remain narrowly focused and less engaged.

6 Limitations and future directions

This study was based on data from uniquely structured groups, which may limit the generalisability of the findings to other contexts. In addition, small group sizes could affect result reliability. Further research is necessary to evaluate the applicability and robustness of the coding scheme across diverse organised group designs with varying study elements.

Our analysis revealed that identical group designs and similar participant demographics do not guarantee consistent group engagement. Understanding what makes group interaction successful and what measures can be used to evaluate it is a critical area of future work. This includes research into specific message types, e.g., self-disclosure or gratitude, that can enhance group engagement [12], analysing which messages receive replies and determining the topics that generate the longest threads [21, 53]. It is important to study participation styles and to identify influential members [7]. Thoughtfully designed prompts are also crucial for promoting effective support exchange among group members [39]. Future research needs to explore these aspects further, particularly through longitudinal studies.

Having similar interactions with Prompts across both cohorts suggests the need for comparative studies of groups with and without Prompts to assess their impact on engagement. In addition, contrasting self-organised and organised groups could shed light on variations in the type of support (e.g., professional vs. peer) and overall effectiveness (e.g., time to achieve goals). A within-group study design could offer more controlled insights into group engagement and support mechanisms to address variability in personality traits that can affect interaction and outcomes in between-group settings. Future studies could also explore how to design environments that balance structure and spontaneity, encouraging meaningful discussions and organic interactions. For example, using CAs, guided by our coding scheme in Section 4.2, with potential functionalities outlined in Table 6.

Table 6: Proposed CA functions.

Category		CA Function Description
Bid	Bids for Inter-action	1. CA requests support from the whole group based on a participant's message that did not get any response.
	Unprompted Support	2. CA takes a proactive approach by sending unprompted messages with relevant information to the group and check-in questions to less active members.
Response	Turning Toward	3. CA responds to messages left without replies from other participants.
	Turning Against	4. CA can acknowledge the message and redirect the conversation to a positive topic.
	Turning Away	5. CA identifies and reacts to the absence of the response, using strategies 1-3.
Social Presence		Use social presence cues to enhance OPSGs' supportive environment.

7 Conclusions

We presented the analysis of chat messages from two cohorts of the MOV'D randomised controlled trial aimed at reducing sedentary behaviour, one characterized by a supportive environment and the other by a lack thereof, and developed a coding scheme. Despite similar conditions, group engagement and quality of interactions varied greatly, suggesting group composition and organic interactions may play a larger role than OPSG design. Sharing personal stories encouraged reciprocity and mutual support. Social Presence cues compensated for the lack of nonverbal communication, shaping the group's social environment. Diversity in support types, whether provided or requested, and peer support quality depended on group characteristics. In sum, this paper advances our understanding of the role structured elements, such as prompts and tasks, play in OPSGs and highlights the significance of organic interactions in fostering engagement and support. We encourage future research to explore further how OPSG design can be optimised to promote organic interactions and create more supportive and dynamic group environments to enhance the effectiveness of OPSGs.

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A MOV'D Study components

All study components with their description are listed in Table 7. More information can be found in the protocol paper [34]. Figure 3 illustrates the change over time in interaction with study components. BCT videos were available only on Sundays, which explains why participants discussed them primarily on Sunday and Monday. In contrast, Prompts and Snackactivities were intended for engagement every weekday, so participants were expected to discuss them accordingly. While both cohorts interacted with all three structured elements, Cohort 3 showed noticeably higher engagement with Snackactivities.

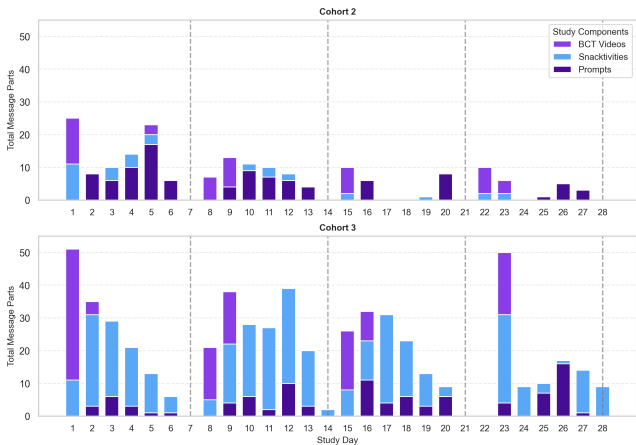


Figure 3: Change over time in message parts by study components: Prompts, Snackactivities (including snackactivity videos, daily progress, and goal setting), and BCT videos. Vertical lines indicate the end of each week, with weeks running from Sunday to Saturday.

B Network charts of Cohort 2 and Cohort 3

A network analysis of the cohorts reveals an unequal distribution of messages. In the network charts, nodes represent participants, directed edges represent messages, the same node colours indicate buddies and node numbers are participants' IDs, see Figure 4.

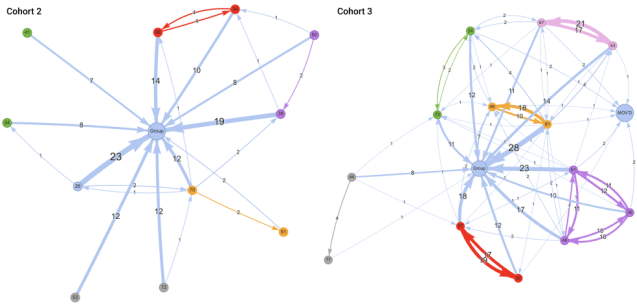


Figure 4: Network graphs of Cohort 2 and Cohort 3.

C Examples of Supportive Messages

Table 8 presents examples and definitions for each support type.

Table 7: MOV'D Study components.

Study component	Frequency	Description
Behaviour change technique videos	Once a week, on Sundays	Overall, participants received four sets of behaviour change videos once per week, with each set focusing on a different aspect of habit formation. As part of the study, each participant was expected to choose one of the videos to watch and then summarize it to their buddy in the group chat, utilizing the learning-by-teaching method to reinforce their understanding of the content [17, 26].
Snackitivity videos	Every weekday	The snackitivity videos were two to five minutes long and undertaking the activities didn't require any special training or any special equipment, except for some snackivities that involved a sturdy desk and/or chair.
Daily progress	Every weekday	Participants shared their exercise plans for the day and their progress.
Goal setting	Once a week, on Sundays	Goal setting entailed writing a message with a week's goal in the group chat.
Intervention prompts	Every weekday	Intervention prompts are messages designed to reinforce the exercise snack behaviours and behaviour change strategies taught in the intervention [9, 26].

Table 8: Examples and definitions of supportive messages.

Type	Definition	Requested Support	Provided Support	
		Bid for Interaction	Unprompted Support	Turning Toward
Esteem Support	Messages that help to promote one's skills, abilities, and intrinsic value, often through compliments or validation [13].	<i>"it's hard to get activity in on meeting heavy days. I've gotten three snackivities in but they are a bit boring—just walking or stairs."</i>	<i>"Thanks everyone! It's been so motivating to hear your tips/tricks/progress."</i>	<i>"I am impressed by your creativity: Simon says, steps ups during hall duty... I get inspired to find opportunities to sneak activities in during mundane tasks."</i>
Informational Support	Messages that include advice, facts or feedback on actions [13].	<i>"Anyone having issue with their Fitbit not syncing correctly?"</i>	<i>"I did this video this morning for all the folks looking for more dance workouts"</i>	<i>"Are the zone minutes actually being recorded on your Fitbit? If not, it may be too loose on your wrist..."</i>
Network Support	Messages that help to enhance one's sense of belonging among people with similar interests and concerns [13].	<i>No such messages in our dataset.</i>	<i>"@P77 What's your goal for this week?"</i>	<i>"Love to know how well it works for you "</i>
Emotional Support	Messages that include caring, concern, encouragement, empathy, and sympathy [13].	<i>"Love the video and does look like fun. But is there an easier one for those of us who are move challenged 🤔"</i>	<i>"I hope everyone is able to meet your snackitivity goals today! I'm going to stay active along with you!"</i>	<i>"@P46 you are not dancing in front of people, so just follow the moves as much as you can. No one will judge you for not doing the exact moves. 🤔"</i>